

PROGAE : FULL-TRACK MUSIC REPRESENTATION LEARNING WITH LATENT TRAJECTORY COMPRESSION

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ABSTRACT

Music representation models operate on short audio excerpts (3–30 s), yet songs span several minutes with rich temporal structure—evolving mood arcs, harmonic progressions, structural builds—that average pooling of chunk embeddings discards. We present a self-supervised framework for learning a compact *temporal token* for full-length recordings by compressing a sequence of frozen pre-trained audio embeddings through a variational bottleneck. The central hypothesis is that temporal evolution can be represented by a single low-dimensional embedding that is complementary to, rather than redundant with, the standard mean-pooled content token. To evaluate this hypothesis without requiring dense musical annotations, we introduce the *Music Semantic Progress Function* (MSPF), a closed-form descriptor of the cumulative rate of semantic change across a recording. We evaluate whether the temporal token reconstructs embedding trajectories, responds to order-changing perturbations, exposes structure-related information to lightweight probes, and remains distinct from the content token. These experiments establish the temporal token as a compact representation for analysis, retrieval, and future generative conditioning.

1. INTRODUCTION

[Introduction: to be written.]

2. RELATED WORK

[Related Work: to be written.]

3. METHOD

3.1. Problem Formulation

Let $\mathcal{E}$ be a frozen pre-trained audio representation model with receptive field T_s s. Given a full-length track of duration T_a , we apply $\mathcal{E}$ with stride Δ and obtain a sequence of local music embeddings

$$\mathbf{z} = (z_1, z_2, \dots, z_T), \quad z_t \in \mathbb{R}^D, \quad T \approx \frac{T_a - T_s}{\Delta}. \quad (1)$$

Each z_t summarizes the local musical content around time t , but the full sequence $\mathbf{z}$ contains additional information about how the track evolves: structural repetitions, builds, drops, sectional boundaries, and long-range semantic arcs. The standard track-level representation is the temporal mean

$$z_{[c]} = \frac{1}{T} \sum_{t=1}^T z_t, \quad z_{[c]} \in \mathbb{R}^D, \quad (2)$$

which aggregates local content but is order-invariant. Any permutation of $\mathbf{z}$ produces the same $z_{[c]}$, so average pooling cannot distinguish, for example, a track that gradually builds into a chorus from one with similar semantics that begins with high intensity and decays.

Our goal is to learn a compact temporal token

$$z_{[\tau]} \in \mathbb{R}^d, \quad d \ll TD, \quad (3)$$

that captures the temporal evolution of the embedding trajectory while remaining complementary to the content token $z_{[c]}$. We train a self-supervised autoencoder over frozen embedding trajectories: an aggregator compresses $\mathbf{z}$ into $z_{[\tau]}$, and a conditional Transformer decoder reconstructs the original residual or velocity trajectory from $z_{[\tau]}$, positional decoder inputs, and an anchor embedding z_a .

3.2. Content Anchor and Reconstruction Targets

To prevent the temporal token from having to encode absolute track-level content, the decoder is provided with an anchor embedding z_a . We consider two anchors:

$$z_a = z_1 \quad (\text{start-frame anchor}), \quad z_a = z_{[c]} \quad (\text{content anchor}). \quad (4)$$

The anchor defines the content coordinate system in which the temporal token describes a trajectory.

We train the model to reconstruct either residuals or velocities. The residual target is

$$r_t = z_t - z_a, \quad t = 1, \dots, T, \quad (5)$$

while the velocity target is

$$v_t = z_{t+1} - z_t, \quad t = 1, \dots, T-1. \quad (6)$$

Residual prediction directly reconstructs the trajectory relative to the anchor. Velocity prediction instead reconstructs local semantic displacement and recovers the sequence by integration:

$$\hat{z}_1 = z_a, \quad \hat{z}_{t+1} = \hat{z}_t + \hat{v}_t. \quad (7)$$

3.3. Temporal Aggregator

The temporal aggregator $\mathcal{A}$ maps the full sequence of frozen embeddings to a pre-bottleneck temporal summary. By default, we prepend a learnable CLS token $[CLS] \in \mathbb{R}^D$ to the sequence and process the result with a lightweight Transformer encoder:

$$[z_{[\tau]}, h_1, \dots, h_T] = \mathcal{A}([CLS, z_1, \dots, z_T]), \quad (8)$$

The aggregator uses Pre-LayerNorm Transformer blocks, multi-head self-attention, GELU/SwiGLU feed-forward layers, and RoPE positional encoding.

We optionally adopt a variational autoencoder formulation, where the temporal token is parametrized as a Gaussian posterior $\mathcal{N}(\mu_\tau, \text{diag}(\sigma_\tau^2))$. The token is sampled using the reparameterisation trick:

$$z_{[\tau]} = \mu_\tau + \sigma_\tau \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I_d). \quad (9)$$

At inference time, unless otherwise stated, we use the posterior mean μ_τ as the deterministic temporal representation for retrieval, probing, and perturbation evaluation.

This aggressive compression is motivated by recent work on long-term motion embeddings, where increasing temporal compression improves latent semantic structure and downstream generation quality [1]. In our setting, we test whether a similar principle holds for music: strong compression should discard local content and noise while preserving the large-scale temporal arc.

3.4. RoPE Mask-Token Transformer Decoder

The decoder reconstructs a full residual or velocity trajectory from a sequence of positional decoder inputs. We initialize a learned mask token $m \in \mathbb{R}^{D_D}$ at each target timestep and add rotary temporal position information:

$$q_t^{(0)} = m + \text{PE}_{\text{RoPE}}(t), \quad t = 1, \dots, T', \quad (10)$$

where $T' = T$ for residual or frame prediction, and $T' = T - 1$ for velocity prediction. These positional inputs specify *where* in the track the decoder should reconstruct. The temporal token $z_{[\tau]}$ specifies *which* temporal trajectory should unfold.

The temporal token is injected into each decoder block using adaptive LayerNorm conditioning. For hidden state x in a decoder block,

$$\text{AdaLN}(x; z_{[\tau]}) = \gamma(z_{[\tau]}) \odot \text{LN}(x) + \beta(z_{[\tau]}), \quad (11)$$

where $\gamma(\cdot)$ and $\beta(\cdot)$ are learned projections from $z_{[\tau]}$. In the default AdaLN-Zero parameterisation, the modulation layers are initialized such that the decoder begins near an identity or weakly conditioned function, stabilising early training.

The decoder is therefore a conditional temporal generator:

$$\hat{\mathbf{v}} = \mathcal{D}_\theta \left(\{m + \text{PE}_{\text{RoPE}}(t)\}_{t=1}^{T-1}; z_{[\tau]}, z_a \right), \quad (12)$$

for velocity prediction, or

$$\hat{\mathbf{r}} = \mathcal{D}_\theta \left(\{m + \text{PE}_{\text{RoPE}}(t)\}_{t=1}^T; z_{[\tau]}, z_a \right), \quad (13)$$

for residual prediction. The anchor z_a is supplied either by concatenating a projected anchor embedding to each positional input, by using a separate AdaLN modulation, or by cross-attending to an anchor token; we use AdaLN by default and ablate other approaches.

This design differs from a standard sequence autoencoder in two ways. First, the decoder never receives the original input tokens; it starts from position-aware mask tokens, forcing the trajectory to be generated from the compressed temporal token. Second, the decoder reconstructs relative motion in a content coordinate system defined by z_a , encouraging $z_{[\tau]}$ to encode temporal evolution rather than static content.

3.5. Encoder-Side Corruption and Span Masking

Decoder mask tokens alone do not prevent the aggregator from seeing the full input sequence. To further encourage global temporal

abstraction, we optionally corrupt the encoder input with temporal masking:

$$\tilde{\mathbf{z}} = \text{Mask}(\mathbf{z}; \mathcal{M}), \quad z_{[\tau]} = \mathcal{A}(\tilde{\mathbf{z}}), \quad (14)$$

where $\mathcal{M}$ is a set of masked timesteps. We consider two masking schemes.

Random timestep masking masks isolated timesteps sampled uniformly across the track. This tests robustness to missing local observations but may be too easy because neighbouring music embeddings are highly correlated.

Contiguous span masking masks temporally contiguous spans of length L_{span} . This is the default corruption strategy for the masked-input ablation because it forces the aggregator to infer section-level evolution from incomplete context. We sweep

$$p_{\text{mask}} \in \{0, 0.25, 0.5, 0.75\}, \quad L_{\text{span}} \in \{5, 15, 30, 60\} \text{ s}. \quad (15)$$

The decoder always reconstructs the full residual or velocity trajectory, including masked regions.

3.6. Training Objective

The default training objective combines velocity reconstruction and KL regularisation:

$$\begin{aligned} \mathcal{L} = & \lambda_v \underbrace{\frac{1}{T-1} \sum_{t=1}^{T-1} \|\hat{\mathbf{v}}_t - (z_{t+1} - z_t)\|_2^2}_{\mathcal{L}_{\text{vel}}} \\ & + \beta \underbrace{D_{\text{KL}}(\mathcal{N}(\mu_\tau, \sigma_\tau^2) \parallel \mathcal{N}(0, I))}_{\mathcal{L}_{\text{KL}}}. \end{aligned} \quad (16)$$

When training residual-prediction models, we replace $\mathcal{L}_{\text{vel}}$ with

$$\mathcal{L}_{\text{res}} = \frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{r}}_t - (z_t - z_a)\|_2^2. \quad (17)$$

For direct frame prediction, we use

$$\mathcal{L}_{\text{frame}} = \frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{z}}_t - z_t\|_2^2. \quad (18)$$

The default model uses velocity prediction with $\lambda_v = 1$, $\lambda_r = \lambda_m = \lambda_{\text{dis}} = 0$, and $\beta = 0.01$.

3.7. Training Details

Unless otherwise stated, models are trained on MTG-Jamendo [2] using frozen MuQ embeddings [3] extracted at 1 Hz. We use AdamW with learning rate 10^{-4} , cosine decay, 5 000 warmup steps, batch size 64, and train for 100 epochs. The default aggregator is a 4-layer Transformer encoder with 8 attention heads and feed-forward dimension 2048. The default decoder is a RoPE-indexed Transformer decoder with AdaLN conditioning from $z_{[\tau]}$. Unless otherwise specified, $d = 4$, $\beta = 0.01$, $\Delta = 1$ s, and the start-frame anchor $z_a = z_1$ are used.

4. EXPERIMENTS

We evaluate whether a compact temporal token captures the semantic evolution of full-length music beyond order-invariant content pooling.

Unless otherwise stated, all models are trained on MTG-Jamendo and evaluated on held-out tracks. The main experiments use frozen

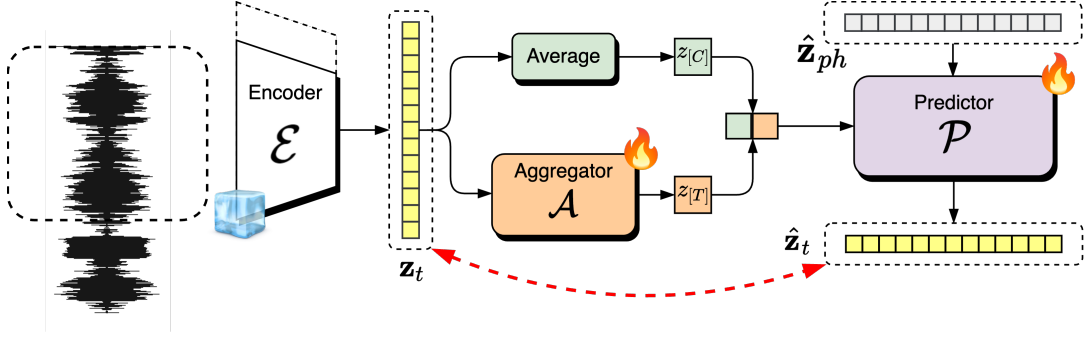


Fig. 1. Overview of the proposed approach for learning dynamic information in a compressed representation. A pretrained audio encoder $\mathcal{E}$ produces an embedding sequence $\mathbf{z}$; the temporal aggregator $\mathcal{A}$ compresses this into a temporal token $z_{[T]}$, while the content token $z_{[C]}$ is computed as a mean over time. The predictor $\mathcal{P}$ reconstructs the original sequence $\hat{\mathbf{z}}$ from learned positional decoder inputs, conditioned on both $z_{[C]}$ and $z_{[T]}$. The red arrow denotes the training objective between $\hat{\mathbf{z}}$ and $\mathbf{z}$.

MuQ embeddings extracted at 1 Hz. Retrieval and distance-based experiments use cosine distance for the content token $z_{[C]}$ and Euclidean distance for the temporal token $z_{[T]}$ after standardisation.

The evaluation contract is fixed across the suite. Reconstruction, MSPF, perturbation, retrieval, and compression results use the same held-out split and are reported at the track level. Controlled perturbations are generated from the evaluation audio before feature extraction, while exact permutation tests are applied directly to the frozen embedding sequence. Unless otherwise stated, MSPF metrics are computed in MuQ space; cross-encoder MSPF checks recompute the same metrics with MERT, USAD, and Music2Latent features. Diffusion experiments draw N samples for each conditioning pair and report both best-sample and distributional statistics so that stochastic generation is not evaluated from a single cherry-picked trajectory.

4.1. Datasets and Evaluation Settings

We use MTG-Jamendo as the primary training corpus because it contains full-length music recordings at scale. Held-out MTG-Jamendo tracks are used for trajectory reconstruction, MSPF-based evaluation, perturbation sensitivity, and ablations. For supervised validation of musical structure, we use datasets with functional segment annotations, such as SALAMI and Harmonix. These annotations provide section boundaries and function labels such as intro, verse, chorus, bridge, instrumental, outro, and silence.

For structure probing, we follow the standard formulation of music structure analysis as two frame-level tasks: boundary detection and function prediction. Boundary detection identifies section changes, while function prediction assigns a structural role to each frame. Following prior work, we report boundary F-scores at 0.5 s and 3 s tolerance, denoted HR.5F and HR3F, and frame-level function prediction metrics, including pairwise F-score (PWF) and accuracy (ACC).

Figure 2 provides the qualitative MSPF validation used throughout the experimental section. It visualises MSPF curves alongside musical structure annotations and controlled temporal variants.

4.2. Trajectory Reconstruction

The first experiment tests whether the compressed temporal token preserves enough information to reconstruct the trajectory of frozen music embeddings. Given an input sequence $\mathbf{z}$, the model encodes

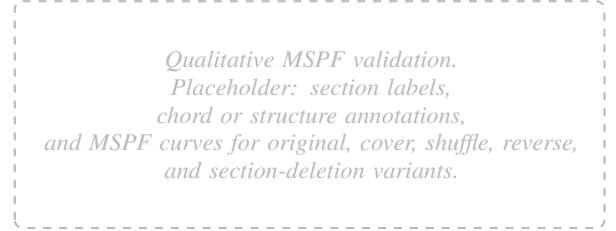


Fig. 2. Qualitative validation of the Music Semantic Progress Function (MSPF). The figure illustrates how MSPF evolves over time for an original track and controlled temporal variants. The purpose is to verify that MSPF follows large-scale musical change while reacting to transformations that alter temporal order.

it into $z_{[T]}$ and reconstructs the trajectory $\hat{\mathbf{z}}$ using the decoder conditioned on $z_{[T]}$ and an anchor z_a . This evaluates whether a single low-dimensional token can represent the temporal evolution that is discarded by order-invariant pooling.

We compare temporal-token models against content-only baselines. The content-only decoder receives the same anchor information but no temporal token, and therefore measures how much of the trajectory can be reconstructed from static content alone. We additionally include a model trained to reconstruct from content-only conditioning, which serves as a collapse diagnostic: if temporal information is unavailable, the decoder should converge toward an average or content-dominated trajectory.

We report two metrics. First, per-frame reconstruction error measures fidelity to the original embedding sequence:

$$\mathcal{E}_{\text{rec}} = \frac{1}{T} \sum_{t=1}^T \|\hat{z}_t - z_t\|_2^2. \quad (19)$$

Second, MSPF R^2 measures how much variance in semantic progress is preserved by the reconstructed trajectory:

$$R_{\text{MSPF}}^2 = 1 - \frac{\sum_t (\hat{S}_t - \hat{\bar{S}})^2}{\sum_t (\hat{S}_t - \bar{S})^2}. \quad (20)$$

The combination of these metrics separates local reconstruction quality from preservation of the global semantic trajectory. Table 1 reports

Table 1. Trajectory reconstruction on held-out tracks. Lower reconstruction error and higher MSPF R^2 indicate better preservation of the embedding trajectory and its semantic-progress profile.

Model	$\mathcal{E}_{\text{rec}} \downarrow$	MSPF $R^2 \uparrow$
Content mean $z_{[C]}$	--	--
Content-only trained decoder	--	--
$z_{[\tau]}, d = 1$	--	--
$z_{[\tau]}, d = 4$	--	--
$z_{[\tau]}, d = 16$	--	--
$z_{[\tau]}, d = 64$	--	--

Table 2. MSPF validation under controlled transformations. DTW distance measures disagreement between the MSPF of the original track and the transformed version.

Condition	MSPF DTW $\downarrow$
Cover version	--
Pitch shift	--
Moderate time stretch	--
Gain change	--
	MSPF DTW $\uparrow$
Chunk shuffle	--
Reverse	--
Section deletion	--

trajectory reconstruction for content-only baselines and temporal-token models with different bottleneck dimensions.

4.3. Annotation-Free Validation of MSPF

The MSPF is intended to provide an annotation-free descriptor of temporal semantic progress. Before using it as a reference signal, we validate its behaviour under controlled transformations. A useful temporal-progress descriptor should remain similar when the large-scale musical arc is preserved, but should change when the ordering or structure of the track is disrupted.

We compare the MSPF of an original track to MSPFs computed from several variants: cover versions, random chunk shuffles, reversals, and section deletions. Covers are used as qualitative or case-study examples because they may preserve large-scale temporal form while varying timbre, production, tempo, or key. Shuffling, reversal, and section deletion provide stronger controlled tests because they explicitly alter temporal order or remove structural material.

We report dynamic time warping distance between MSPF curves:

$$d_{\text{MSPF}}(i, j) = \text{DTW}(\hat{S}(\mathbf{z}_i), \hat{S}(\mathbf{z}_j)). \quad (21)$$

DTW is used because musically corresponding events may occur at slightly different times, especially for covers or time-stretched variants. For transformations where absolute timing is important, we additionally visualise the unwarped MSPF curves in Figure 2. Table 2 summarises the MSPF-DTW distances for transformations expected to preserve or disrupt temporal progress.

After validating MSPF as a temporal-progress descriptor, we test whether the learned temporal-token geometry agrees with MSPF-based similarity. We compute pairwise temporal-token distances and compare them to pairwise MSPF-DTW distances over held-out tracks.

Table 3. Agreement between representation geometry and MSPF-based temporal-progress distance.

Representation	Spearman $\rho_{\tau, \text{MSPF}} \uparrow$
Random	--
Content token $z_{[C]}$	--
MSPF feature baseline	--
Temporal token $z_{[\tau]}$	--
$[z_{[C]}, z_{[\tau]}]$	--

Table 4. Cross-encoder MSPF validation. The temporal token is trained from MuQ embeddings; MSPF metrics are recomputed using an independent scoring encoder.

MSPF encoder	MSPF $R^2 \uparrow$	$\rho_{\tau, \text{MSPF}} \uparrow$	$\rho_{\text{sep}} \uparrow$
MuQ	--	--	--
MERT	--	--	--
USAD	--	--	--
Music2Latent	--	--	--

Because retrieval and neighbourhood structure depend on ranking rather than calibrated distances, we report Spearman correlation:

$$\rho_{\tau, \text{MSPF}} = \text{corr}_{\text{rank}} \left(\|z_{[\tau]}^{(i)} - z_{[\tau]}^{(j)}\|_2, d_{\text{MSPF}}(i, j) \right). \quad (22)$$

This experiment asks whether tracks that are close in temporal-token space also have similar semantic-progress profiles. Table 3 reports this agreement for the content token, MSPF features, the temporal token, and their combination.

To reduce circularity, we repeat the MSPF validation with encoders that are not used to train the default temporal token. For each held-out track we compute the default $z_{[\tau]}$ from MuQ, then compute MSPF curves and MSPF-DTW distances in an independent frozen encoder space. Agreement under this protocol indicates that the temporal token is not merely matching idiosyncrasies of the MuQ feature geometry. Table 4 reports this cross-encoder MSPF validation.

4.4. Temporal Retrieval Case Studies

We complement the quantitative MSPF-geometry test with qualitative nearest-neighbour retrieval. Given a query track, we retrieve nearest neighbours using only $z_{[C]}$, only $z_{[\tau]}$, and the concatenated representation $[z_{[C]}, z_{[\tau]}]$. The expected behaviour is different for each space: $z_{[C]}$ should favour similar timbre, genre, and instrumentation, while $z_{[\tau]}$ should favour similar long-range evolution such as gradual build-up, drop-and-return form, or repeated verse-chorus alternation. The concatenated representation is used as a practical retrieval mode when both static identity and temporal shape matter.

Figure 3 visualises representative queries by plotting the query MSPF curve beside the nearest-neighbour curves returned by each representation. Table 5 summarises the same case studies with compact musical descriptors.

4.5. Sensitivity to Temporal Perturbations

A temporal representation should be stable under transformations that preserve musical evolution, while responding to edits that change temporal order or remove structural material. We therefore construct two families of perturbations. The preserving family includes pitch

*Temporal retrieval case studies.
Placeholder: query MSPF and
nearest-neighbour MSPFs from
 $z_{[c]}$, $z_{[\tau]}$, and $[z_{[c]}, z_{[\tau]}]$ retrieval.*

Fig. 3. Qualitative temporal retrieval. The comparison checks whether $z_{[\tau]}$ retrieves tracks with similar semantic-progress curves even when content-token retrieval favours more similar surface audio characteristics.

Table 5. Nearest-neighbour retrieval case studies. Entries are qualitative descriptors of the top retrieved track under each representation.

Query	$z_{[c]}$ NN	$z_{[\tau]}$ NN	$[z_{[c]}, z_{[\tau]}]$ NN
Gradual build	--	--	--
Verse-chorus	--	--	--
Drop-return	--	--	--
Sparse outro	--	--	--

shift, moderate time stretch, gain change, and optional generative edits such as instrument insertion when the global arrangement is preserved. The disrupting family includes chunk shuffle, reversal, and section deletion.

For each original track x and perturbed track x' , we measure normalised displacement in temporal-token space:

$$\delta_{\tau}(x, x') = \frac{\|z_{[\tau]}(x) - z_{[\tau]}(x')\|_2}{\bar{d}_{\tau}}, \quad (23)$$

where $\bar{d}_{\tau}$ is the mean pairwise temporal-token distance over the evaluation set. We summarise the separation between disrupting and preserving perturbations as

$$\rho_{\text{sep}} = \frac{\mathbb{E}_{x, x' \in \mathcal{D}_{\text{disrupt}}}[\delta_{\tau}(x, x')]}{\mathbb{E}_{x, x' \in \mathcal{D}_{\text{preserve}}}[\delta_{\tau}(x, x')]} \quad (24)$$

This ratio gives a compact measure of whether the representation distinguishes temporal edits from nuisance changes.

For chunk shuffling, we additionally define a continuous severity score based on the amount of temporal displacement induced by the permutation π :

$$s(\pi) = \frac{\sum_{t=1}^T |t - \pi(t)|}{\max_{\pi'} \sum_{t=1}^T |t - \pi'(t)|}. \quad (25)$$

We then report the Spearman correlation between shuffle severity and temporal-token displacement. This tests whether the token changes gradually with the magnitude of the temporal edit rather than merely detecting that a transformation has occurred. Figure 4 visualises this relation, Table 6 reports displacement under preserving and disrupting perturbations, and Table 7 reports the shuffle-severity correlation.

4.6. Structure Probing

We next test whether the temporal token contains information useful for annotated music structure analysis. Instead of retrieving tracks

*Perturbation sensitivity.
Placeholder: temporal-token dis-
placement versus shuffle severity,
with preserving transforma-
tions shown as reference points.*

Fig. 4. Sensitivity of the temporal token to controlled perturbations. The experiment measures whether displacement in temporal-token space increases with the severity of temporal reordering while remaining small for preserving transformations.

Table 6. Perturbation sensitivity. Preserving transformations should produce small displacement, while disrupting transformations should produce larger displacement.

Rep.	Pitch↓	Stretch↓	Gain↓	Disrupt↑	$\rho_{\text{sep}} \uparrow$
$z_{[c]}$	--	--	--	--	--
MSPF feat.	--	--	--	--	--
$z_{[\tau]}, d = 1$	--	--	--	--	--
$z_{[\tau]}, d = 4$	--	--	--	--	--
$z_{[\tau]}, d = 16$	--	--	--	--	--
$z_{[\tau]}, d = 64$	--	--	--	--	--

with similar symbolic form, which can be sparse and ill-defined for small annotated datasets, we evaluate boundary detection and function prediction using lightweight probes.

The probing setup is intentionally simple. We freeze all representations and train a linear frame-level head. For boundary detection, the head predicts whether each frame corresponds to a section change. For function prediction, it predicts one of the structural function labels. The probe receives local frame features, global tokens, or their concatenation:

$$u_t \in \{z_t, [z_t, z_{[c]}], [z_t, z_{[\tau]}], [z_t, z_{[c]}, z_{[\tau]}], [z_{[\tau]}, \text{PE}(t)], [z_{[c]}, \text{PE}(t)]\}. \quad (26)$$

Including positional encoding allows global tokens to be used for frame-level prediction. Comparing $[z_t, z_{[\tau]}]$ to z_t tests whether the temporal token provides global form information beyond local frame features. Comparing $z_{[\tau]}$ to $z_{[c]}$ tests whether the temporal token is more informative than order-invariant content for structure-related prediction.

Boundary detection is evaluated with HR.5F and HR3F. Function prediction is evaluated with PWF and ACC:

$$\mathcal{L}_{\text{probe}} = \mathcal{L}_{\text{BCE}}^{\text{boundary}} + \mathcal{L}_{\text{CE}}^{\text{function}}. \quad (27)$$

The probe is not intended to establish state-of-the-art music structure analysis. Its purpose is to test whether the frozen representations linearly expose information about structural boundaries and functions. Table 8 reports the probing results for local features, global content and temporal tokens, and their concatenations.

4.7. Content-Temporal Disentanglement

We evaluate whether $z_{[\tau]}$ captures temporal order information not available to mean pooling. The first test uses exact sequence permutations. Given an embedding sequence $\mathbf{z}$, we construct permuted

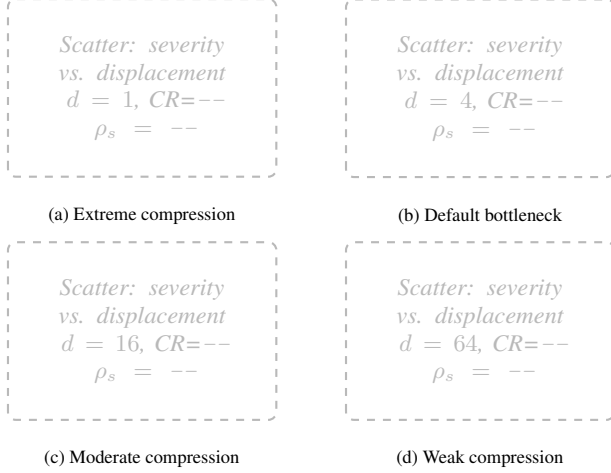


Fig. 5. Relationship between temporal perturbation severity and temporal-token displacement for different compression ratios. Each point corresponds to a chunk-shuffled track, with severity computed from Eq. (25) and displacement computed from Eq. (23). The Spearman correlations shown in each panel correspond to the values reported in Table 7.

Table 7. Correlation between chunk-shuffle severity and temporal-token displacement.

Representation	Spearman $\rho(s, \delta_\tau) \uparrow$
$z_{[c]}$	--
MSPF feat.	--
$z_{[\tau]}, d = 1$	--
$z_{[\tau]}, d = 4$	--
$z_{[\tau]}, d = 16$	--
$z_{[\tau]}, d = 64$	--

variants $\pi(\mathbf{z})$ using random shuffle, reversal, half-swap, circular shift, and local shuffle. These transformations preserve the multiset of frame embeddings, and therefore leave the mean-pooled content token unchanged up to numerical precision:

$$\Delta_C = \|z_{[c]}(\mathbf{z}) - z_{[c]}(\pi(\mathbf{z}))\|_2, \quad \Delta_\tau = \|z_{[\tau]}(\mathbf{z}) - z_{[\tau]}(\pi(\mathbf{z}))\|_2. \quad (28)$$

A representation of temporal evolution should produce larger Δ_τ for order-altering transformations, while Δ_C serves as an order-invariant reference. Table 9 reports the content-token and temporal-token displacements under exact sequence permutations.

Finally, we quantify similarity between content and temporal representations using representation-level dependence measures. We report linear CKA or CKNNA between $z_{[\tau]}$ and $z_{[c]}$, together with two references: a same-representation upper reference and a shuffled-pair lower reference. These values are used comparatively rather than as absolute estimates of independence. Table 10 reports these dependence measures across reference comparisons and temporal-token variants.

4.8. Generative Conditioning and Token Swapping

The strongest test of complementarity is generative: if $z_{[c]}$ and $z_{[\tau]}$ separate content from temporal evolution, then a conditional generator should follow the content source when $z_{[c]}$ is swapped and the

Table 8. Linear probing for music structure analysis. Boundary detection is evaluated using HR.5F and HR3F; function prediction is evaluated using PWF and ACC.

Probe input	HR.5F $\uparrow$	HR3F $\uparrow$	PWF $\uparrow$	ACC $\uparrow$
Local sequence z_t	--	--	--	--
$[z_t, z_{[c]}]$	--	--	--	--
$[z_t, z_{[\tau]}]$	--	--	--	--
$[z_t, z_{[c]}, z_{[\tau]}]$	--	--	--	--
$[z_{[c]}, \text{PE}(t)]$	--	--	--	--
$[z_{[\tau]}, \text{PE}(t)]$	--	--	--	--

Table 9. Order sensitivity under exact sequence permutations. Mean pooling is invariant to permutations, while a temporal token should respond to order changes.

Transform	$\Delta_C \downarrow$	$\Delta_\tau \uparrow$
Random shuffle	--	--
Reverse	--	--
Half-swap	--	--
Circular shift	--	--
Local shuffle	--	--

temporal source when $z_{[\tau]}$ is swapped. We therefore train a small diffusion model in the frozen embedding-trajectory space. The model denoises a trajectory while conditioning on both global tokens:

$$\tilde{\mathbf{z}} = G_\phi(\epsilon; z_{[c]}, z_{[\tau]}), \quad \epsilon \sim \mathcal{N}(0, I). \quad (29)$$

At test time, we decode matched and swapped conditions from two tracks A and B :

$$(z_{[c]}(A), z_{[\tau]}(A)), \quad (z_{[c]}(B), z_{[\tau]}(B)), \\ (z_{[c]}(A), z_{[\tau]}(B)), \quad (z_{[c]}(B), z_{[\tau]}(A)). \quad (30)$$

For each condition c we draw N generated trajectories $\{\tilde{\mathbf{z}}_c^{(n)}\}_{n=1}^N$. Content alignment is measured by comparing the mean generated embedding $\bar{\tilde{\mathbf{z}}}_c^{(n)}$ to the requested content token in $z_{[c]}$ -space. Temporal alignment is measured by comparing the MSPF of each generated trajectory to the MSPF of the requested temporal source; an additional temporal-token alignment is computed by re-encoding $\tilde{\mathbf{z}}_c^{(n)}$ with A . We report best-of- N reconstruction error, mean reconstruction error, sample diversity, and minimum/mean MSPF error:

$$\mathcal{E}_{\text{best}}(c) = \min_n \mathcal{E}_{\text{rec}}(\tilde{\mathbf{z}}_c^{(n)}, \mathbf{z}_{\text{target}}), \\ \mathcal{E}_{\text{div}}(c) = \frac{2}{N(N-1)} \sum_{i < j} \text{dist}(\tilde{\mathbf{z}}_c^{(i)}, \tilde{\mathbf{z}}_c^{(j)}). \quad (31)$$

This experiment directly measures whether $z_{[c]}$ controls static musical identity while $z_{[\tau]}$ controls the evolution of the generated track, and whether the conditional generator covers a distribution of plausible trajectories rather than a single averaged path. Table 11 reports the full distributional swap-alignment protocol.

4.9. Compression Summary

Compression is a central experimental axis, but the main paper uses it only to choose and justify a single-token operating point. We sweep

Table 10. Representation dependence between temporal and content tokens. Shuffled pairs provide a lower reference; identical or content-like representations provide an upper reference.

Comparison	CKA↓	CKNNA↓
$z_{[c]}$ vs. $z_{[c]}$	--	--
$z_{[\tau]}$ vs. shuffled $z_{[c]}$	--	--
$z_{[\tau]}$ vs. $z_{[c]}$	--	--
$z_{[\tau]}$ vs. $z_{[c]}$, no anchor	--	--
$z_{[\tau]}$ vs. $z_{[c]}$, VAE bottleneck	--	--

the temporal bottleneck dimension d while keeping one temporal token. The compression ratio is

$$\text{CR} = \frac{TD}{d}. \quad (32)$$

Table 12 summarises the trade-off between trajectory fidelity, temporal utility, and content leakage. To make the compression comparison independent of optimisation budget, we also report a compute-matched sweep in which each bottleneck is trained under both a fixed-update budget and a fixed wall-clock budget. The training-objective, anchor/target, masking, and multi-token capacity ablations are reported in the appendix because they refine the design rather than establish the core claim that a single $z_{[\tau]}$ can encode track-level temporal evolution.

4.10. Encoder Receptive Field

The frozen encoder defines the local units from which the temporal token is built. We therefore evaluate whether encoder receptive field and frame rate affect temporal-content disentanglement. We train the same temporal bottleneck over multiple frozen encoders with different temporal resolutions and context lengths. Shorter receptive fields provide more local observations, while longer receptive fields may already mix local content and temporal context before the aggregator.

For each encoder, we report temporal utility and content leakage using the same downstream metrics as above. This experiment tests whether the quality of $z_{[\tau]}$ depends primarily on the bottleneck architecture or also on the temporal granularity of the frozen features. Table 13 reports the effect of frozen encoder receptive field and frame rate.

5. REFERENCES

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Table 11. Diffusion token swapping. A successful split should make generated content follow $z_{[C]}$ and generated temporal evolution follow $z_{[\tau]}$, including under crossed conditions. Best and mean errors are computed over N samples per condition.

Condition	C align. $\uparrow$	τ align. $\uparrow$	Best err. $\downarrow$	Mean err. $\downarrow$	Div. $\uparrow$	MSPF min $\downarrow$	MSPF mean $\downarrow$
$(z_{[C]}(A), z_{[\tau]}(A))$	--	--	--	--	--	--	--
$(z_{[C]}(B), z_{[\tau]}(B))$	--	--	--	--	--	--	--
$(z_{[C]}(A), z_{[\tau]}(B))$	--	--	--	--	--	--	--
$(z_{[C]}(B), z_{[\tau]}(A))$	--	--	--	--	--	--	--

Table 12. Single-token compression sweep. The default model uses $d = 4$, selected to balance temporal utility and separation from the content token. Fixed-update and fixed-time columns provide compute-matched checks for the compression trend.

d	CR	$\mathcal{E}_{\text{rec}} \downarrow$	MSPF $R^2 \uparrow$	$\rho_{\text{sep}} \uparrow$	$\text{CKA}(z_{[\tau]}, z_{[C]}) \downarrow$	Fixed upd. $\uparrow$	Fixed time $\uparrow$
1	--	--	--	--	--	--	--
4	--	--	--	--	--	--	--
16	--	--	--	--	--	--	--
64	--	--	--	--	--	--	--
256	--	--	--	--	--	--	--

Table 13. Effect of frozen encoder receptive field and frame rate. The same temporal-token model is trained on each encoder.

Encoder	RF	MSPF $R^2 \uparrow$	$\rho_{\text{sep}} \uparrow$	$\text{CKA}(z_{[\tau]}, z_{[C]}) \downarrow$
MuQ	~ 10 s	--	--	--
MERT	~ 10 s	--	--	--
USAD	~ 1 s	--	--	--
Music2Latent	var.	--	--	--
MATPAC	~ 1 s	--	--	--